

Table 1: Few-shot BABILong ablation results for `tiiuae/falcon-mamba-7b-instruct`. Accuracy is exact-match accuracy (%) averaged over QA1–QA3 with 50 examples per task and context length. The no-cache row is imported from the reasoning-curves approximate next-token/prefix baseline; StateCache rows use the few-shot BABILong scaffold with inserted `[CACHE]` tokens and first-token candidate scoring.

Method	0k	1k	2k	4k	8k	16k	Avg.
No cache, few-shot	52.0	36.7	28.0	24.7	5.3	0.0	24.4
Loss-selected, $k = 1$, L62	28.0	33.3	34.0	16.7	18.7	18.0	24.8
Loss-selected, $k = 4$, L62	29.3	34.7	36.7	14.7	22.7	18.7	26.1
Loss-selected, $k = 8$, L62	27.3	40.0	37.3	18.7	17.3	20.7	26.9
Loss-selected, $k = 4$, L62, train up to 8k	25.3	30.7	30.0	34.0	33.3	24.0	29.6
Loss-selected, $k = 4$, L62, no gate	32.0	36.0	37.3	20.7	18.7	24.0	28.1
Loss-selected, $k = 4$, L32	48.7	43.3	36.7	27.3	30.7	26.0	35.4
Uniform interval, $k = 4$, L62	31.3	31.3	38.7	19.3	22.0	18.0	26.8
Random, $k = 4$, L62	27.3	21.3	21.3	18.0	22.0	13.3	20.6
Regex/entity, L62	28.7	28.7	36.0	18.0	13.3	21.3	24.3